

SATQUERY AI — REMOTE SENSING ANALYSIS REPORT

ISRO Problem Statement 26167 • Mission ID: 0defe715 • Generated: 17:53:30.904Z

Natural Query:	Highlight the surrounding water bodies.
Classified Task:	GROUNDING
Calibrated Confidence:	90% (Spatial Intersection & IoU Match)
Specialist Models:	satquery-grounding-rs-v1
Total Latency:	120.1 ms

GROUNDING AI INTERPRETATION

Successfully grounded 'water body'. Identified 2 prominent region(s) spanning a total surface area of 233.4 hectares (2.334 km²), centered at approximately 13.725046° N, 80.205562° E (8.04% of the satellite footprint).
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INPUT SENSOR METADATA

Filename	Modality	Dimensions	Bands	GeoTIFF CRS
isro_sdsc_spaceport.	OPTICAL	512x512	3	EPSG:4326

DELINEATED VECTOR TARGETS & SPATIAL TELEMETRY

Feature	Area (ha)	Area (km²)	Perimeter (m)	Centroid [Lat, Lon]	Confidence
Water body #1	197.80	1.9780	11969.7	13.725046°, 80.205562°	82%
Water body #2	35.60	0.3560	2396.5	13.733643°, 80.225635°	80%

OBSERVABLE AGENTIC EXECUTION TRACE

#	Phase / Event	Observable Details	Time (m)
1	Conversational Memory Loaded (Turn #2)	Retrieved session 'session-isro-demo-2' with 2 prior messages. Context conditioned on conversation state.	0.0
2	Query Ingestion	Received natural language query: 'Highlight the surrounding water bodies.' with 1 uploaded image(s).	0.0
3	Input Validation	Verified 1 image file(s). Format and size limits satisfied.	0.0
4	Raster Metadata Extraction: isro_sdsc_spaceport.tif	Extracted dimensions (512x512), 3 band(s). CRS: EPSG:4326	55.7
5	Modality Identification	Detected sensor modalities: ['OPTICAL'].	0.0
6	Query Classification → GROUNDING	Single image detected with spatial referring/localization intent -> Routed to Text-Guided Grounding Specialist.	0.0

7	Model Dispatch	Dispatched task to specialist: 'satquery-grounding-rs-v1' on device 'cpu'.	0.0
8	Specialist Execution Completed	Inference finalized in 63.7ms. Evidence artifacts compiled.	63.7
9	Result Fusion & Calibration	Derived grounded response with calibrated confidence 90% (Spatial Intersection & IoU Match).	0.0